

scam-shield

Production-Grade Real-Time Payment Fraud Defence

The bet no bank has been willing to make: put AI where it cannot do regulatory harm.

< 350ms

End-to-End Decision

Dedicated

Private Link Cluster

9 SQL

Flink Statements

3 LLMs

Gemini·GPT-4o·Granite



PRODUCER

HMAC-SHA256

IBM Stack · Confluent Cloud · HashiCorp Terraform · Azure · RBIH MuleHunter.AI · Google Gemini · IBM WatsonX



KAFKA

Private Link



FLINK ★

9 SQL



DECIDE

350ms



GOVERNED AI

Narrate

₹36,014 Cr

Lost to bank fraud in a single year.

Source: RBI Annual Report FY 2024-25

The problem is accelerating — not slowing. Every year without production-grade defence is another ₹36,000 crore.

Sources: RBI Annual Report FY25 · RBI H1 FY25 · Ministry of Finance Lok Sabha 2025

194%

Surge in fraud
value FY24-25

8×

Jump in banking
fraud H1 FY25

1.13M

Cyber fraud cases
registered 2023

56%

UPI share of all
digital fraud

WHY IT STAYS UNSOLVED

Technology isn't the gap. The architecture is.



Speed vs Accuracy

UPI clears in milliseconds. Batch ML scoring is too slow. Banks choose speed — and accept more fraud.



PII + AI = Regulatory Trap

Sending customer data to cloud AI violates RBI data localisation. Banks can't use powerful AI safely.



Black-Box Decisions

ML models cannot explain a BLOCK to an RBI examiner. No explainability means no regulatory defensibility.



Siloed Payment Rails

Fraud jumps UPI → IMPS → Card in seconds. Per-rail tools built in isolation miss coordinated attacks.



No Human-in-the-Loop at Scale

Thousands of daily alerts bury the most dangerous cases. Analysts cannot review what they cannot find.

THE ARCHITECTURE BET

Flink decides. AI explains.

The governance architecture that makes streaming AI safe to deploy in a regulated bank — not deploying a model, deploying the framework that makes it defensible.

INV-1

No raw PII ever reaches a model

INV-2

Every model call is governed & audited

INV-3

Decision is deterministic & replayable

INV-4

All topics follow domain.entity.event

Three Zones · One Decision Boundary



fraud.alert.raised → single boundary crossing · PII-free · post-decision · immutable

Tier-1 P99 < 120ms (hard-block / Flink-only) · Tier-2 P99 < 350ms (full ensemble)

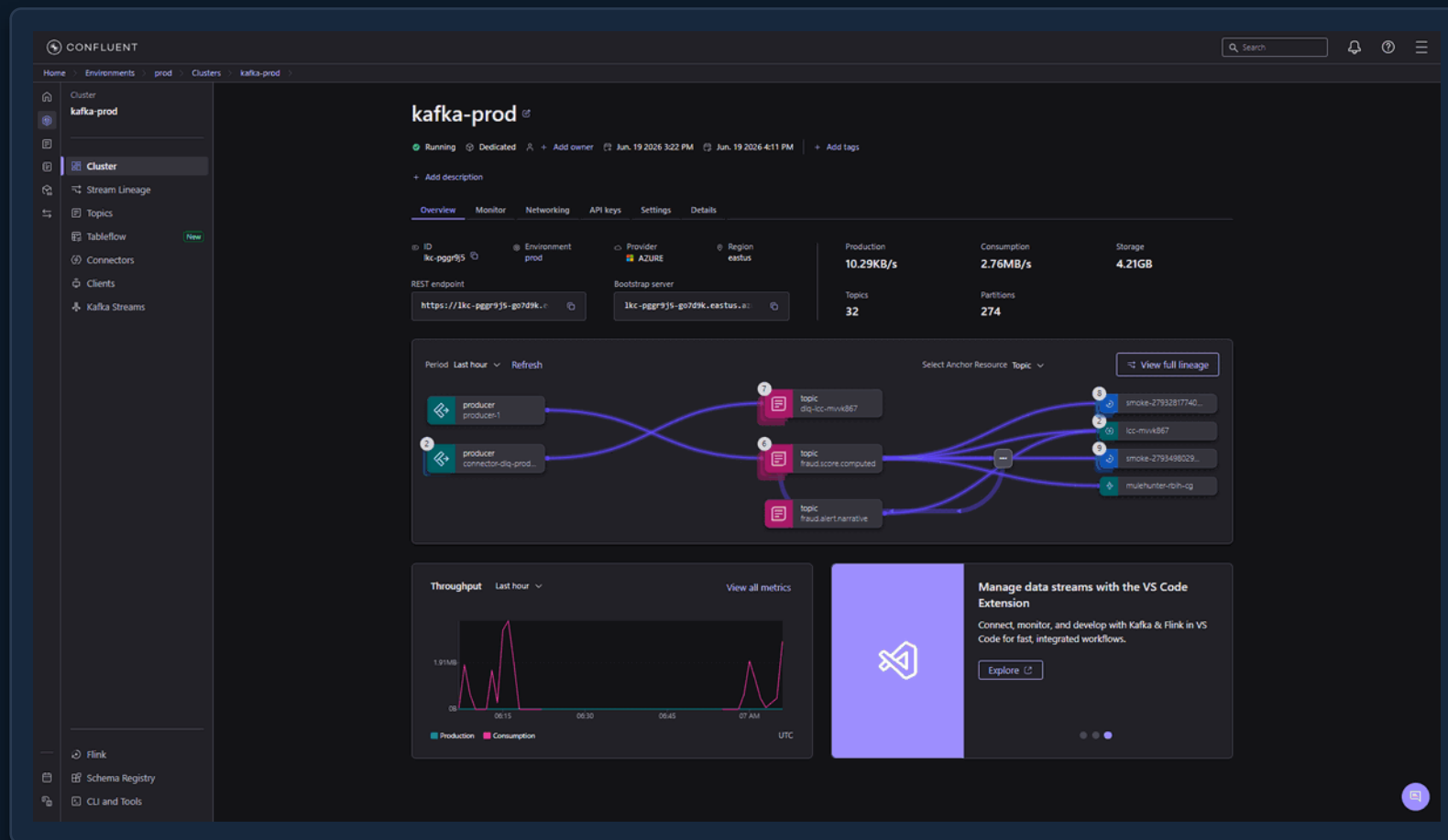
A dedicated, private cluster on Azure East US

LIVE

WHAT YOU'RE SEEING

- Dedicated cluster on Private Link, status Running
- Provider Azure · region eastus · 274 partitions
- 32 topics, ~4.2 GB stored, live throughput shown
- Stream Lineage maps producers → topics → consumers

WHY IT MATTERS Enterprise-grade isolation with an auditable, replayable event backbone — not a demo cluster.



9 SQL Statements. 15 Logical Jobs. The Deterministic Detection Core.

< 120ms

Tier-1 P99

< 350ms

Tier-2 P99

9

Statements

15

Logical Jobs

~40%

CFU saved

●	STMT-INGEST	Rail filter · dedup · sanity check	JOB-01/02/03
●	STMT-ENRICH	11 temporal joins · velocity · behavioral	JOB-04
●	STMT-RULES	Per-rail CEP · MATCH_RECOGNIZE	JOB-05/06/07
●	STMT-CEPXR	Cross-rail CEP · ATO + Mule chain	JOB-08
●	STMT-DECIDE ★	350ms fusion · 40/40/20 · hard-block first	JOB-09/10/12
●	STMT-COMPLAINT	30-day interval join · preventive HOLD	JOB-11
●	STMT-NARRATE ★	ML_PREDICT → Gemini / GPT-4o / WatsonX	AI overlay

SCORE FUSION

ensemble =

Flink Rules × 40%

MuleHunter.AI × 40%

AML Engine × 20%

Decision window: 350ms tumbling
Hard-block fires FIRST — bypasses ensemble
Tier-1 fast path: < 120ms P99
Tier-2 full: < 350ms P99

The nine SQL statements, running live

WHAT YOU'RE SEEING

- ingest → enrich → rules → cepxr → decide → narrate
- plus stmt-publish, stmt-intel and stmt-complaint
- all Running · ~1 CFU each · zero messages behind
- one shared, elastically-scaled compute pool

WHY IT MATTERS *The 9 SQL statements from the architecture slide — deployed and observable, not slideware.*

The screenshot shows the Confluent Flink interface. On the left is a sidebar with navigation options: Home, Environments (prod), Environment overview, Clusters, Flink, Streaming agents, Network management, Stream Governance, Schema Registry, Catalog management, Integrations, and Artifacts. The main panel displays the 'Flink' section with a search bar and filters. Below the filters is a table of statements.

☐	Flink statement name	Status	Statement type	Statement CFU	State size GB	Created	# Messages behind total
☐	stmt-decide flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	18 hours ago	0 Not behind
☐	stmt-complaint flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	18 hours ago	0 Not behind
☐	stmt-enrich flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
☐	stmt-ingest flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
☐	stmt-rules flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	-	19 hours ago	0 Not behind
☐	stmt-publish flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	-	19 hours ago	0 Not behind
☐	stmt-intel flink-pool-prod-prod	Running	EXECUTE STATEMENT...	1	0	19 hours ago	0 Not behind
☐	adhoc-describe-2790232 flink-pool-prod-prod	Stopped	SELECT	-	-	20 hours ago	-
☐	adhoc-describe-2789680 flink-pool-prod-prod	Stopped	SELECT	-	-	a day ago	-
☐	smoke-27896559527-vali flink-pool-prod-prod	Stopped	SELECT	-	-	a day ago	-

STMT-CEPXR + STMT-DECIDE · Deterministic. Explainable. Replayable.

STMT-CEPXR — XR_ATO_PROBE_AND_DRAIN

```
SELECT * FROM
  `payments.transaction.enriched`
MATCH_RECOGNIZE (
  PARTITION BY account_id_hash
  ORDER BY `$rowtime`
  MEASURES
    LAST(BIG.txn_id) AS txn_id,
    FIRST(PROBE.`$rowtime`) AS start
  ONE ROW PER MATCH
  AFTER MATCH SKIP TO NEXT ROW
  PATTERN (PROBE BIG)
  WITHIN INTERVAL '5' MINUTE
  DEFINE
    PROBE AS PROBE.amount < 100,
    BIG AS BIG.amount > 10000
    AND BIG.bene_txn_count_30d = 0
);

-- XR_MULE_CHAIN: 3+ credits → debit
PATTERN (CRD{3,} OUTB)
WITHIN INTERVAL '10' MINUTE
DEFINE CRD AS direction='CREDIT',
  OUTB AS direction='DEBIT'
  AND bene_txn_count_30d = 0
```

STMT-DECIDE — Score Fusion (350ms)

```
-- 40/40/20 weighted ensemble
ensemble_score =
  0.40 * flink_rules_score
+ 0.40 * mulehunter_score
+ 0.20 * COALESCE(aml_score,
  behavioral_score, 0.0)

-- 350ms tumbling window
FROM TABLE(TUMBLE(
  TABLE `fraud.score.computed`,
  INTERVAL '0.350' SECOND))

-- Thresholds (priority order)
HARD_BLOCK override → first
rules_score >= 0.80 → BLOCK
ensemble >= 0.70 → BLOCK
ensemble >= 0.60 → CHALLENGE
ensemble >= 0.40 → HOLD
else → ALLOW

-- 6 atomic sinks (STATEMENT SET)
fraud.alert.raised
+ fraud.account.freeze
+ platform.ops.event
```

Three LLM Backends.

One Governed Topic. Zero PII. Post-Decision Only.

INV-2 · every call governed

Google Gemini

PRIMARY

googleai provider

Native Confluent Flink provider. Zero proxy. Publicly reachable from Confluent Cloud.

```
CREATE MODEL `gemini_narrate`
WITH (
  'provider'          = 'googleai',
  'googleai.connection' = 'gemini-conn',
  'googleai.system_prompt' =
    'You are a fraud analyst...'
);
```

```
-- confluent flink connection create
gemini-conn --type googleai
```

INV-3 · runs after DECIDE

Azure OpenAI GPT-4o

SUPPORTED

azureopenai provider

Native Confluent Flink azureopenai provider. Azure auth handled natively — zero proxy.

```
CREATE MODEL `azure_openai_narrate`
WITH (
  'provider'          = 'azureopenai',
  'azureopenai.connection' =
    'azure-openai-conn',
  'azureopenai.system_prompt' =
    'You are a fraud analyst...'
);
```

```
-- confluent flink connection create
azure-openai-conn --type azureopenai
```

Zero PII · post-decision

IBM WatsonX Granite

SUPPORTED

openai-compat proxy

OpenAI-compat proxy on Azure Container Apps. Auto IBM IAM token refresh + project_id injection.

```
CREATE MODEL `watsonx_narrate`
WITH (
  'provider'          = 'openai',
  'openai.connection' = 'watsonx-conn',
  'openai.model_version' =
    'ibm/granite-4-h-small'
);
```

```
-- Proxy on Azure Container Apps
-- IBM IAM auto-refresh (~55 min)
-- project_id injection
```

```
All 3 → ML_PREDICT('model', CONCAT(alert_id, ' | Action: ', final_action, ' | Vuln: ', vulnerability_tier)) →
fraud.alert.narrative · model_id stamps provenance
```

AI narration runs inside Flink — live

WHAT YOU'RE SEEING

- CREATE MODEL gemini_narrate (Google Gemini backend)
- ML_PREDICT invoked straight from Flink SQL
- A HOLD alert → a plain-English analyst note in seconds
- Empathetic, PII-free and stamped with the model id

WHY IT MATTERS *STMT-NARRATE in action — post-decision AI explains the alert; it never makes the decision (INV-3).*

The screenshot displays the Confluent Flink SQL IDE interface. The left sidebar shows a workspace tree with folders for 'examples', 'marketplace', 'prod', 'dev', and 'scam-shield'. The main editor area shows two SQL queries and their execution results.

Query 1:

```
1 CREATE MODEL 'gemini_narrate'
2 INPUT (prompt STRING)
3 OUTPUT (response STRING)
4 WITH (
5   'provider' = 'googleai',
6   'task' = 'text_generation',
7   'googleai.connection' = 'gemini-conn',
8   'googleai.system_prompt' =
9     'You are a fraud analyst. In AT MOST 2 short sentences, explain why the
10    described transaction was flagged with the given final_action. Be factual.
11    Do NOT use the customer''s name or any account identifier.'
12 );
```

Execution Result 1:

workspace-2026-06-23-0814... Completed Jun. 23, 2026, 3:43 PM Mode Streaming Run

Success

Model 'gemini_narrate' with version '2' created

Query 2:

```
1 SELECT t.prompt, p.response
2 FROM (VALUES ('Alert A1 | Action: HOLD | Vulnerability tier: DIGITAL_VULNERABLE | Reason code: APP_NEWBENE_DWELL')) AS t(prompt)
3 CROSS JOIN LATERAL TABLE(ML_PREDICT('gemini_narrate', t.prompt)) AS p;
```

Execution Result 2:

workspace-2026-06-23-0814... Completed Jun. 23, 2026, 3:42 PM 1 Mode Streaming Run

prompt	response
Alert A1 Action: HOLD...	The transaction was placed on hold because a digitally vulnerable customer attempted a transfer to a newly added beneficiary via the mobile app. The system flagge...

Last saved at 6/23/26, 3:43 PM 2 / 20 cells

AI that Assists. Never Decides. Always Audited.

LANGGRAPH — 13 NODES

	orchestrator_node	validate + route
	context_enrichment_node	Qdrant RAG + GPT-4o
	upi / card / transfer nodes	specialist scorers ×3
	pattern_gap_node ★	Learning Loop F3
	decision_node ⚡ HITL	interrupt_before gate
	case_management_node	SAR draft + queue
	notification_node	PMLA §8A templates
	compliance_node	NCRP + RBI DPIP
	sar_generation_node	Multi-signal SAR draft
	knowledge_update_node ★	Learning Loop F1
	output_node	ops.event + gov audit

HITL POLICY

	HARD BLOCK Auto-freeze (any amount) · flag_ncrp always HITL
	BLOCK $\leq ₹5L \geq 0.85$ Auto-freeze + notify · flag_ncrp HITL
	BLOCK $> ₹5L$ Notify only · freeze always HITL
	SAR required ALL actions → HITL · No exceptions
	CHALLENGE ≥ 0.85 Auto send_otp_challenge only
	HOLD any All actions → HITL

4 MCP SERVERS

- fraudscoringmcpserver :8005
- fraudinvestigationmcpserver :8009
- amltriagemcpserver :8010
- fraudactionmcpserver :5002

The Data Contracts That Make It Production-Ready

MANAGED CONNECTORS

PostgresCdcSourceV2

SOURCE

10 compacted lookup topics (account baseline, sanctions, velocity...) ·
Debezium pgoutput · TopicRegexRouter SMT · feeds FOR SYSTEM_TIME AS
OF in STMT-ENRICH

PostgresSink — Alerts

SINK

fraud.alert.raised + fraud.alert.narrative → sinks.fraud_alerts +
fraud_alert_narratives · UPSERT on alert_id · Dashboard AI Analyst Notes
join

PostgresSink — Ops

SINK

platform.ops.event → sinks.ops_events · ReplaceField\$Value SMT drops
nested Avro STRUCT fields before insert

CONFLUENT TABLEFLOW → APACHE ICEBERG

Zero ETL · 4 analytical topics materialised
to open Iceberg lakehouse
1 declarative config replaces 4 sink connectors

**fraud.features.inference**

→ ML Feature Store / AI Inference

**payments.graph.edges**

→ Graph DB staging + analytics

**fraud.intelligence.dpip**

→ NPCI DPIP regulatory export

**fraud.intelligence.mulehunter**

→ MuleHunter retraining corpus

```
tableflow_topics = {  
  "fraud.features.inference" = {  
    enabled=true, table_formats=["ICEBERG"]  
  }  
}
```

SCHEMA

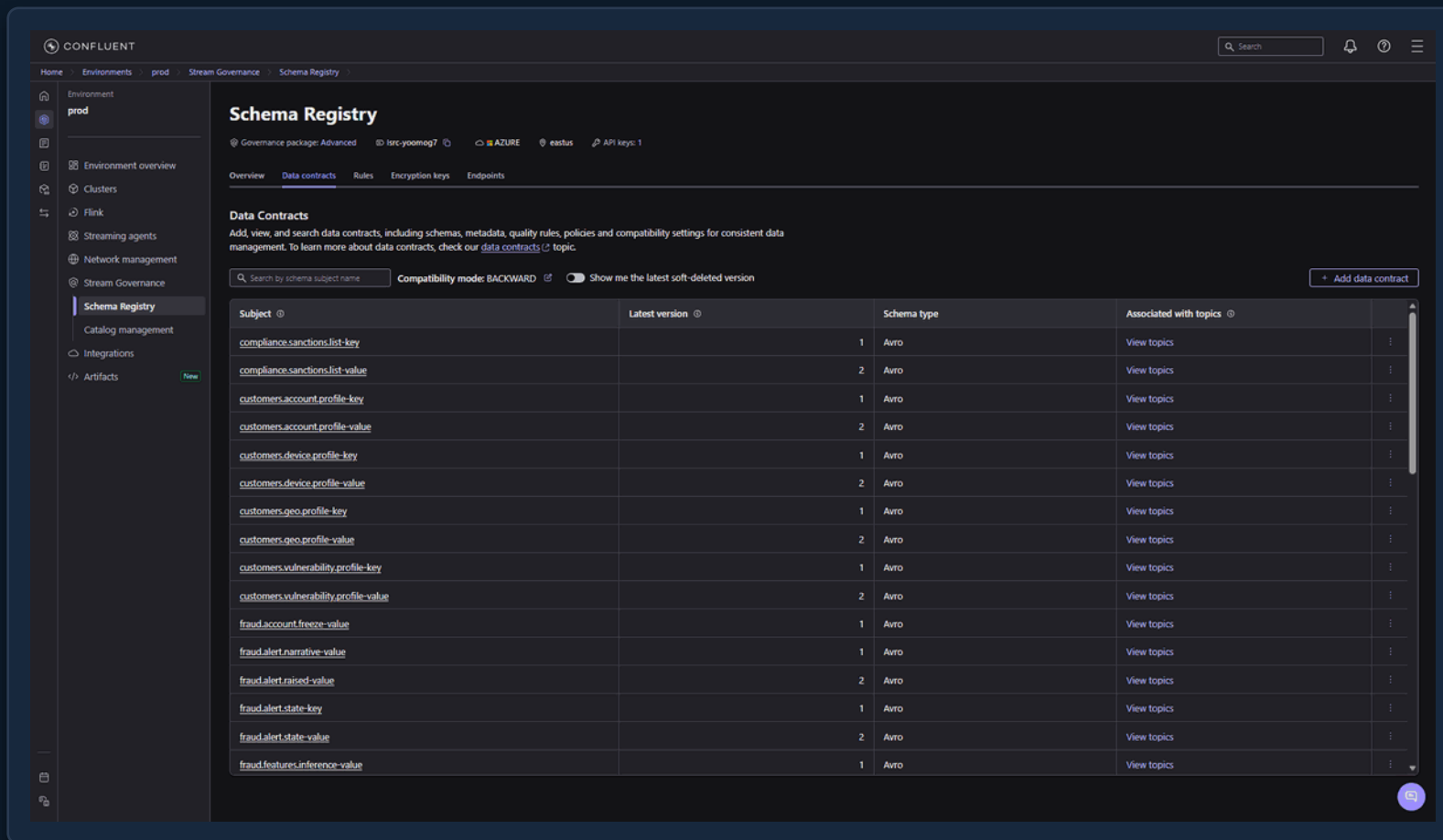
Every topic is an Avro contract

• LIVE

WHAT YOU'RE SEEING

- Subjects registered for both keys and values
- BACKWARD compatibility enforced on every evolution
- Lookup, alert and intelligence schemas all present
- Schema Registry links each contract to its stream

WHY IT MATTERS *No producer can publish a change that silently breaks a consumer — contracts are enforced, not hoped for.*



The screenshot displays the Confluent Schema Registry web interface. The left sidebar shows the navigation menu with 'Schema Registry' selected. The main panel is titled 'Schema Registry' and shows the 'Data contracts' tab. Below the header, there's a search bar and a table of registered subjects. The table has columns for 'Subject', 'Latest version', 'Schema type', and 'Associated with topics'. The subjects listed are all Avro schemas, including compliance, customer profiles, device profiles, geo profiles, vulnerability profiles, fraud alerts, and fraud features. Each subject has a 'View topics' link in the 'Associated with topics' column.

Subject	Latest version	Schema type	Associated with topics
compliance.sanctions.list-key	1	Avro	View topics
compliance.sanctions.list-value	2	Avro	View topics
customers.account.profile-key	1	Avro	View topics
customers.account.profile-value	2	Avro	View topics
customers.device.profile-key	1	Avro	View topics
customers.device.profile-value	2	Avro	View topics
customers.geo.profile-key	1	Avro	View topics
customers.geo.profile-value	2	Avro	View topics
customers.vulnerability.profile-key	1	Avro	View topics
customers.vulnerability.profile-value	2	Avro	View topics
fraud.account.freeze-value	1	Avro	View topics
fraud.alert.narrative-value	1	Avro	View topics
fraud.alert.raised-value	2	Avro	View topics
fraud.alert.state-key	1	Avro	View topics
fraud.alert.state-value	2	Avro	View topics
fraud.features.inference-value	1	Avro	View topics

Enterprise Infrastructure. Not a Demo Cluster.



Confluent Cloud

The Hero — Kafka + Flink

Dedicated cluster · Private Link · Schema Registry · Stream Governance · 9 Flink SQL · Tableflow → Apache Iceberg



HashiCorp Terraform

Infrastructure as Code

Two isolated roots · Zero UI resources · Topic naming enforced at plan time · Least-privilege RBAC · Dual-slot key rotation



Microsoft Azure

Private Cloud Foundation

Private Link data plane · Key Vault (HMAC PII key) · Azure OpenAI GPT-4o · Postgres Flex Server · Container Apps



IBM watsonx / Bob

AI-Assisted Engineering

6 design docs, 9 Flink statements, full IaC, Java producer — architected conversationally. AI operated the platform.



RBIH MuleHunter.AI

Cross-Bank Mule Intelligence

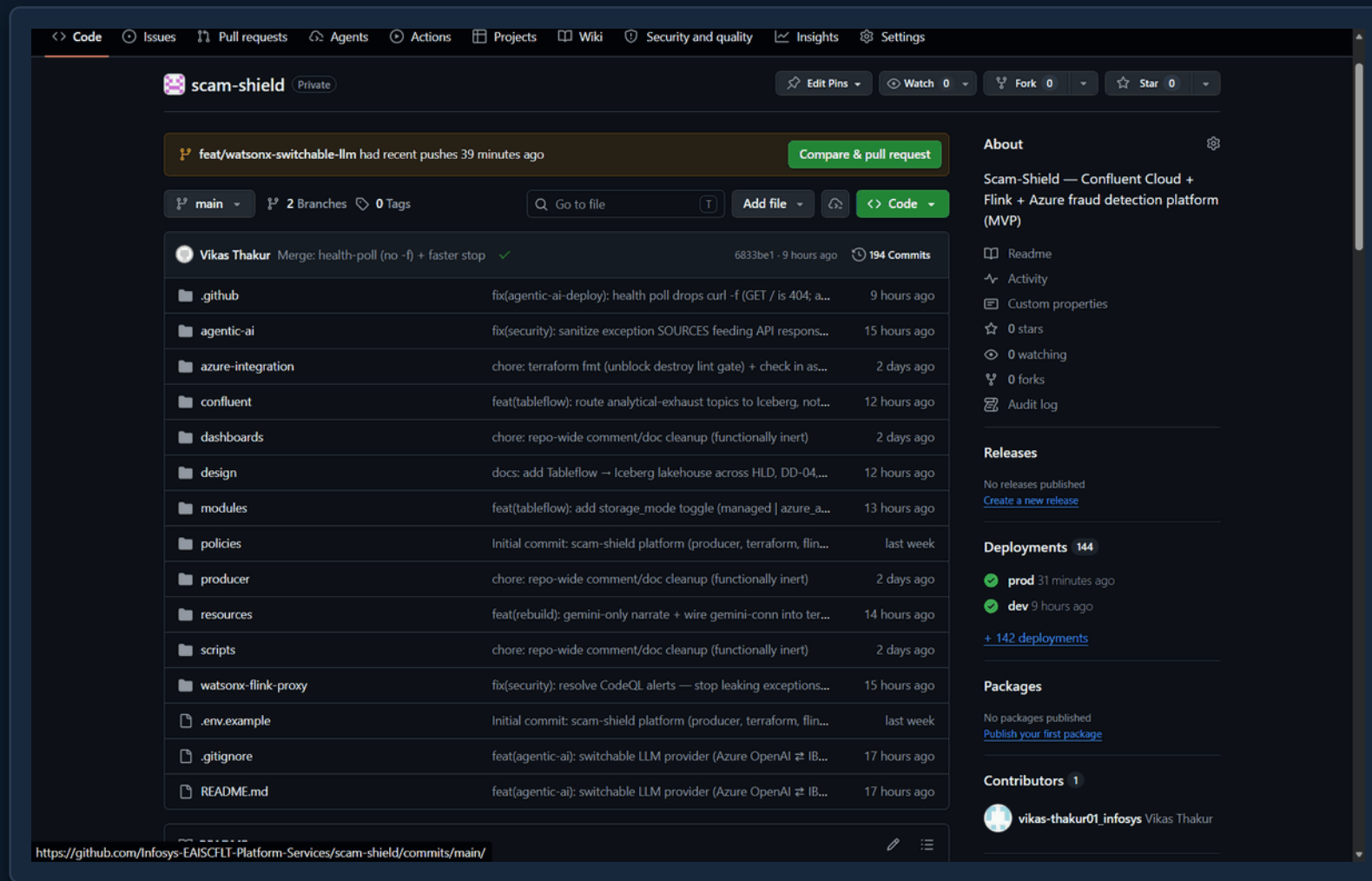
Pure Kafka event contract — no REST, no shared DB · 90-day rolling state · consumer group mulehunter-rbih-cg · 40% weight

Everything is code, in one repo

WHAT YOU'RE SEEING

- One repo holds both roots: confluent/ and azure-integration/
- modules/, resources/ and policies/ sit beside app code
- No resource is ever created by hand in a console
- Drift is reconciled back to Git on every run

WHY IT MATTERS *Terraform is the sole source of truth — if it's not in Git, it doesn't exist.*



One pipeline provisions Confluent and Azure

WHAT YOU'RE SEEING

- confluent/ applies first; azure-integration/ reads its state
- Key Vault, PrivateLink, Postgres and monitoring provisioned
- Apply consumes the approved plan with -auto-approve
- Outputs hand off cleanly between the two roots

WHY IT MATTERS *The whole platform — both clouds — stands up from code in one flow.*

Summary

All jobs

- Resolve Environments
- Lint & Validate — confluent
- Lint & Validate — azure-integration
- Lint & Validate — aws-integration
- Plan — `{{ matrix.environment }}`
- Destroy — `{{ matrix.environment }}`
- Drift Detection — prod
- Plan Azure — prod
- Seed Postgres DDL — `{{ matrix.environment }}`
- Destroy Azure — `{{ matrix.environment }}`
- Plan AWS — `{{ matrix.environment }}`
- Destroy AWS — `{{ matrix.environment }}`
- Apply — `{{ matrix.environment }}`
- Apply Azure — prod**
- Apply AWS — `{{ matrix.environment }}`

Run details

- Usage
- Workflow file

Apply Azure — prod
succeeded 25 minutes ago in 11m 59s

Search logs

Terraform Apply — azure-integration 11m 47s

```

127 azurerm_private_dns_a_record.ccn_wildcard[0]: Creation complete after 1s [id=/subscriptions/***/resourceGroups/rg-confluent-
128 prod/providers/Microsoft.Network/privateDnsZones/g0do07.eastus.azure.confluent.cloud/A/*]
129 |
130 | Warning: Applied changes may be incomplete
131 |
132 | The plan was created with the -target option in effect, so some changes
133 | requested in the configuration may have been ignored and the output values
134 | may not be fully updated. Run the following command to verify that no other
135 | changes are pending:
136 |   terraform plan
137 |
138 | Note that the -target option is not suitable for routine use, and is
139 | provided only for exceptional situations such as recovering from errors or
140 | mistakes, or when Terraform specifically suggests to use it as part of an
141 | error message.
142 |
143 Apply complete! Resources: 7 added, 0 changed, 5 destroyed.
144
145 Outputs:
146
147 action_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/actionGroups/ag-confluent-prod"
148 ccn_private_endpoint_ip = "100.78.217.29"
149 confluent_network_dns_domain = "g0do07.eastus.azure.confluent.cloud"
150 confluent_network_id = "n-g0do07"
151 hackathon_postgres_admin_password = <sensitive>
152 hackathon_postgres_admin_username = "scamadmin"
153 hackathon_postgres_database_name = "scam_shield"
154 hackathon_postgres_fqdn = "psql-scamshield-prod-ppc.postgres.database.azure.com"
155 hackathon_postgres_server_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.DBforPostgreSQL/flexibleServers/psql-scamshield-prod-ppc"
156 log_analytics_workspace_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.OperationalInsights/workspaces/law-confluent-prod"
157 log_analytics_workspace_name = "law-confluent-prod"
158 resource_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod"
159 workbook_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/workbooks/b4bb9e69-98d1-5aa1-9b74-73afc3ca6bc"
  
```

> Post Run actions/checkout@v4 0s

> Complete job 0s

<https://github.com/Infosys-EAISFLT-Platform-Services/scam-shield/actions/runs/28073978849/job/83114296837>

COST

• LIVE

This enterprise stack runs on free credits

WHAT YOU'RE SEEING

- June 2026 accrued charges: \$0.00
- ~\$560 of usage fully offset by program promos
- Remaining free credits tracked in-console
- Transparent, itemised, per-resource billing

WHY IT MATTERS *Production-grade architecture demonstrated at effectively zero cost — and costed transparently for real planning.*

The screenshot displays the Confluent Billing and payment interface. The left sidebar contains navigation links: Home, Environments, Data Portal, SQL workspaces, Cluster links, and Migration hub (marked as 'New'). The main content area is titled 'Invoice Breakdown' and includes a 'Select invoice range' section with dropdowns for Contract (All), Month (June), and Year (2026). Below this, the 'Current accrued charges' for the period Jun. 1 - Jun. 30, 2026 are listed as \$0.00 USD. A 'Free credits' section shows \$65 / \$75 USD remaining. A detailed table lists usage and various promo applications that offset the charges, resulting in a total accrued charge of \$0.00. A 'Download invoice' button is present. The 'Invoice period details' section shows 'Charges by environment (5)' with a dropdown for 'environment - prod (\$518.66 USD)'. The 'Non-cluster-specific charges' section is partially visible at the bottom.

Resource name	Quantity	Unit price	Total
Usage			\$560.51
Promo applied (CONFLUENTDEV1)			-\$1.00
Promo applied (FLINKJAVA101)			-\$9.51
Promo applied (101CONNECT)			-\$25.00
Promo applied (KAFKA101)			-\$25.00
Promo applied (SCHEMA101)			-\$25.00
Promo applied (101SECURITY)			-\$25.00
Promo applied (DOTNETKAFKA101)			-\$25.00
Promo applied (FLINKSQL25)			-\$25.00
Promo applied (FREETRIAL400)			-\$400.00
Total accrued charges			\$0.00

LIVE DEMO

Real Fraud. Real Decisions. Live on Confluent Cloud.

SIM Swap + APP Fraud

UPI P2P

STMT-RULES fires SIM_SWAP-001. STMT-CEPXR catches BENEFICIARY_ADD → AUTH → PAYMENT in 3s (XR dwell). HITL queued. OTP challenge auto-issued.

CHALLENGE

CNP + Geo-Impossible Travel

Debit Card

CNP fraud + geo-impossible (800km in 4 min) + IMEI mismatch. conf 0.88 ≥ 0.85 + amt ≤ ₹5L → auto-freeze. Card specialist AI dossier.

BLOCK

Mule Fan-In Network

IMPS

MuleHunter CRITICAL (mule_prob 0.91). XR_MULE_CHAIN: 3 rapid CREDITS → DEBIT to new beneficiary in 8 min. Hard override → HOLD. Always HITL.

HOLD

Structuring / AML

NEFT

₹12L · 3 round-amount txns in 4h → CTR breached. Sanctions flag. sar_required=true. SAR auto-generated. ALL actions → HITL.

HARD
BLOCK

★ STMT-NARRATE fires post-decision for all 4 scenarios → AI Analyst Note (Gemini primary) appears on Fraud Ops Dashboard within seconds · Flink decides, AI explains, analyst reviews

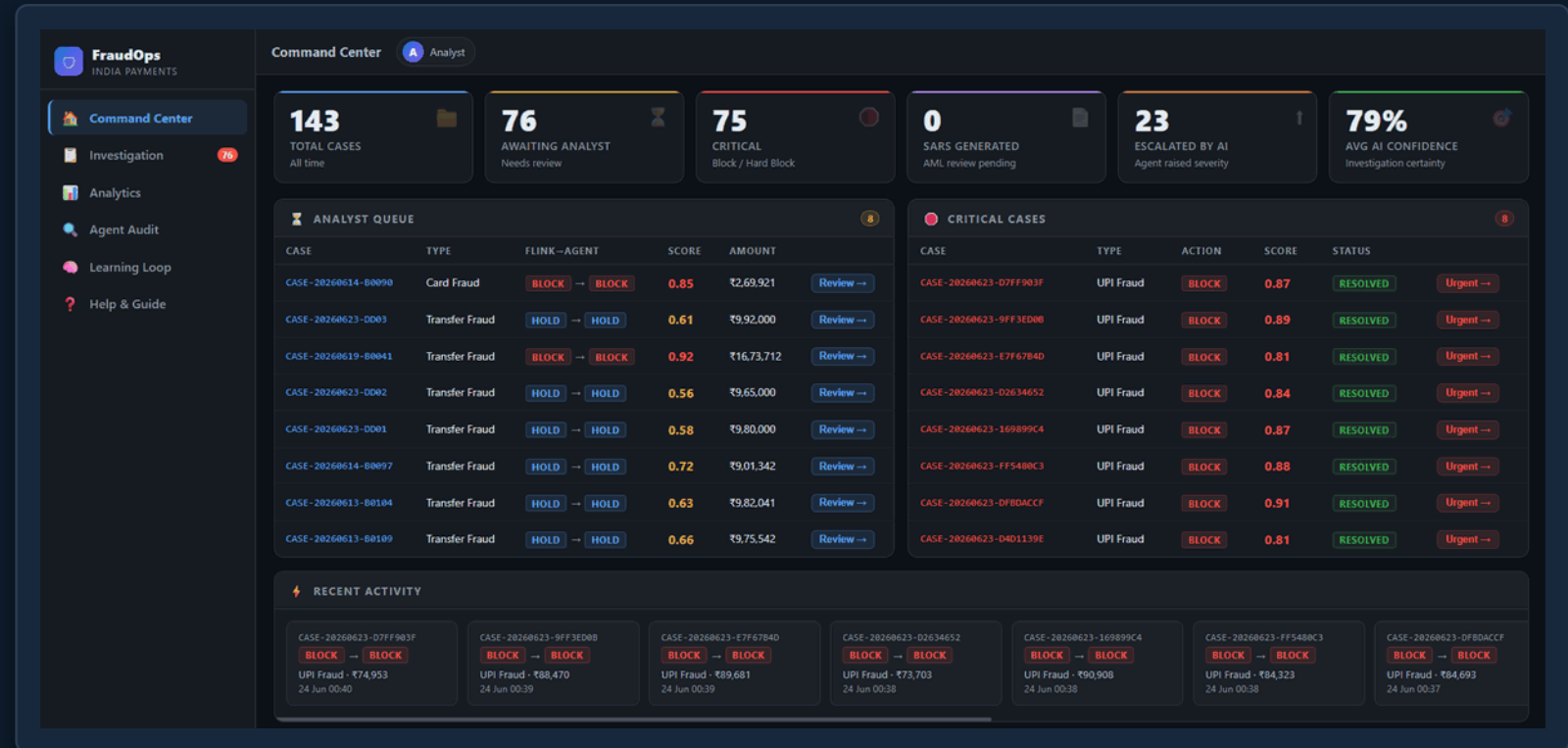
A live command center for fraud ops

• LIVE

WHAT YOU'RE SEEING

- Real-time tiles: total, awaiting analyst, critical, escalated
- 79% of cases auto-handled; the rest queued for a human
- Analyst queue ranked by risk, action and SLA
- Fed by fraud.alert.raised + platform.ops.event over SSE

WHY IT MATTERS *This is what the analyst sees — the whole fraud picture, with the few that matter surfaced first.*



Deterministic score, agent cross-check

WHAT YOU'RE SEEING

- Flink rule score and agent confidence side by side
- Decision severity path from ALLOW through HARD_BLOCK
- Detected fraud patterns and per-rule validation listed
- Python tools cross-validate Flink hits — never re-score

WHY IT MATTERS *The agent corroborates the deterministic decision; it never replaces it (INV-3).*

FraudOps
INDIA PAYMENTS

Investigation Workspace 143

Search case ID, txn, alert...

CASES

BLOCK → BLOCK CRITICAL	CASE-20260623-D7FF903F UPI Fraud 0.87	24 Jun 00:40
BLOCK → BLOCK CRITICAL	CASE-20260623-9FF3ED08 UPI Fraud 0.89	24 Jun 00:39
BLOCK → BLOCK CRITICAL	CASE-20260623-E7F67B4D UPI Fraud 0.81	24 Jun 00:39
BLOCK → BLOCK CRITICAL	CASE-20260623-D2634652 UPI Fraud 0.84	24 Jun 00:38
BLOCK → BLOCK CRITICAL	CASE-20260623-169899C4 UPI Fraud 0.87	24 Jun 00:38
BLOCK → BLOCK CRITICAL	CASE-20260623-FF5480C3 UPI Fraud 0.88	24 Jun 00:38
BLOCK → BLOCK CRITICAL	CASE-20260623-DFB0ACCF UPI Fraud 0.91	24 Jun 00:37
BLOCK → BLOCK CRITICAL	CASE-20260623-D4D1139E UPI Fraud 0.81	24 Jun 00:37
BLOCK → BLOCK CRITICAL	CASE-20260623-2B7F06FF UPI Fraud 0.83	24 Jun 00:36
BLOCK → BLOCK CRITICAL	CASE-20260623-434221A9 UPI Fraud 0.88	24 Jun 00:36

Risk Score Meters

FLINK SCORE 87%

AGENT CONFIDENCE 92%

RISK LEVEL CRITICAL

Decision Severity Path

ALLOW CHALLENGE HOLD **BLOCK F+A** HARD_BLOCK

Fraud Patterns Detected

SIM SWAP TAKEOVER

Rule Validation

SIM-SWAP-001 UNCONFIRMED N/A No additional detail	APP-002 UNCONFIRMED N/A No additional detail
APP-001-TFR UNCONFIRMED N/A No additional detail	VEL-001 ✓ CONFIRMED CRITICAL No additional detail
MULE-001 UNCONFIRMED N/A No additional detail	

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

1. Rules Fired and Their Significance - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.

Every alert gets a plain-English narrative

WHAT YOU'RE SEEING

- A structured investigation narrative is drafted per case
- It cites the specific rules, signals and recommended action
- All model calls routed through the AI Governance gateway
- PII-free — the AI never sees raw identifiers (INV-1)

WHY IT MATTERS Analysts read why, not just a score — exactly the analyst note promised in the pitch.

The screenshot displays the FraudOps India Payments Investigation Workspace. The interface includes a sidebar with navigation options: Command Center, Investigation (76), Analytics, Agent Audit, Learning Loop, and Help & Guide. The main area is titled 'Investigation Workspace' and shows a list of 143 cases. Each case entry includes a status (e.g., BLOCK, CRITICAL), a case ID, a transaction ID, and a risk score. The 'Investigation Narrative' section provides a detailed analysis for a specific case, including 'Investigation Findings' and 'Additional Context'. The findings section lists three points: Rules Fired and Their Significance, Behavioral Signals, and Recommended Action and Confidence Level. The additional context section provides further details about the transaction and the rules triggered.

FraudOps INDIA PAYMENTS

Investigation Workspace 143 Total 76 Awaiting 75 Critical 0 SAR 23 Escalated 67 Resolved Analyst

Overview Investigation SAR Report Actions Log Ops Events

APP-001-TFR UNCONFIRMED N/A No additional detail

VEL-001 ✓ CONFIRMED CRITICAL No additional detail

MULE-001 UNCONFIRMED N/A No additional detail

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

- Rules Fired and Their Significance - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.
- Behavioral Signals - Supporting Signals: - The high behavioral risk score aligns with the VEL-001 rule's findings, suggesting a potential SIM swap takeover attempt. This pattern is consistent with unauthorized access attempts where an attacker intercepts the SIM card to gain control over the UPI account. - The absence of SIM change detection (SIM-SWAP-001) and the lack of new device or first-time beneficiary flags (APP-002) indicate that the account's usual security parameters have not been altered, which is unusual for legitimate transactions of this magnitude. - Contradictory Signals: - The MuleHunter score (0.180) is below the mule threshold (>0.75), suggesting that the account does not exhibit typical mule account behaviors. This lack of corroboration from the MuleHunter model weakens the suspicion of money muling activities, focusing the investigation primarily on the SIM swap risk.
- Recommended Action and Confidence Level - Recommended Action: Initiate a multi-factor authentication (MFA) challenge for the account holder to verify their identity. This could include sending a one-time password (OTP) to the registered mobile number or email, or prompting the account holder to answer security questions. Additionally, monitor the account for any further suspicious activity and consider temporarily restricting high-value transactions until the account's security can be confirmed. - Confidence Level: 0.92. The high confidence level is based on the strong evidence from the VEL-001 rule, which indicates a significant deviation from the account's normal behavior. The behavioral risk score further supports the suspicion of a SIM swap takeover attempt. While the MuleHunter score does not corroborate money muling activities, the overall pattern of evidence strongly suggests that immediate action is necessary to secure the account and prevent potential fraud.

Additional Context

Investigation Findings

The transaction was flagged because it triggered two high-risk rules: 'SIM-SWAP-001' and 'VEL-001', which are associated with the 'SIM_SWAP_UPI_TAKEOVER' pattern. The behavioral risk score is also high, indicating a potential SIM swap takeover attempt, where an attacker may have gained unauthorized access to the customer's UPI account by intercepting the SIM card.

Live 10:23

The platform proposes its own new rules

WHAT YOU'RE SEEING

- Agents cluster resolved cases into emerging fraud patterns
- Each gap becomes a PENDING Flink SQL rule proposal
- An analyst must approve before anything deploys
- Approved rules ship as code via a pull request

WHY IT MATTERS *The system learns and suggests — humans still approve every rule change (INV-3).*

FraudOps
INDIA PAYMENTS

Command Center
Investigation 76
Analytics
Agent Audit
Learning Loop
Help & Guide

Learning Loop
AI-generated rule proposals and emerging fraud pattern clusters. Approve proposals to deploy new Flink rules.

RULE PROPOSALS 0 Refresh

No proposals yet — the agent will surface them as it processes cases.

EMERGING PATTERNS 10 Run Now

UPI < ₹1,000 per txn 10 cases

SIM-Port Account Takeover
10 accounts were compromised following a SIM port attack. Once the attacker controlled the mobile number, they bypassed OTP authentication across banking apps and made transfers (under ₹1,000 each). All 10 cases triggered the same set of detection rules (bc7d3b), suggesting a common attack pattern.
bc7d3b
First seen: 2026-06-23 19:06 Latest: 2026-06-23 19:10 728756d1 Investigate 10 cases →

CARD ₹1,000 – ₹10,000 per txn **NOVEL** 4 cases

Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.
334c4a
First seen: 2026-06-12 00:58 Latest: 2026-06-20 08:58

UPI < ₹1,000 per txn 15 cases

ACCOUNT TAKEOVER UPI
15 UPI cases showing account takeover upi behaviour triggered the same set of detection rules (cf1082), suggesting a common attack pattern.
cf1082
First seen: 2026-06-08 21:58 Latest: 2026-06-21 11:58

TRANSFER ₹1,000 – ₹10,000 per txn 15 cases

Transaction Structuring (Below CTR)
15 TRANSFER transactions were deliberately sized to stay just below the ₹10 lakh reporting threshold (₹1,000–₹10,000 per txn).
First seen: 2026-06-08 21:58 Latest: 2026-06-21 11:58

Novel pattern — Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.

Payment channel	CARD
Transaction amount range	₹1,000 – ₹10,000 per txn
Cases in this group	4 cases
Detection rules triggered	334c4a
First case seen	2026-06-12 00:58
Most recent case	2026-06-20 08:58

Recommended action: Flag for analyst review. Run pattern gap detection. Consider new Flink rule proposal.

Live • 10:26

Not deploying a model.

Deploying the governance architecture that makes it safe.



PII-Safe by Architecture

HMAC-SHA256 at the producer. AI models never see raw PII — not by policy, by infrastructure. INV-1.



AI Never Decides

STMT-DECIDE: Flink rules + MuleHunter + AML. Deterministic, replayable, explainable. INV-3.



3-Backend Governed AI Overlay

STMT-NARRATE: Gemini (primary) / GPT-4o / WatsonX Granite via ML_PREDICT. All audited. INV-2.



Full Terraform IaC + Tableflow

Zero UI resources. Schema Registry. Tableflow → Iceberg. Demo-to-production in one codebase.



₹36,014 Cr Market. RBI-Ready.

PMLA / DPDP Act / INV-1-4. NCRP + DPIIP integrations. 4 audit streams. 10-yr AML retention.

Connectors: PostgresCdcSourceV2 · PostgresSink (alerts) · PostgresSink (ops) | Flink: STMT-DECIDE + STMT-NARRATE | Schema: fraud.score.computed (Avro) | Lineage: raw→validated→enriched→scored→alert→narrative